misses the particlelike aspects demonstrated by measurements that find localized particles that can be, for example, recorded by a single dot on a screen. (See Figure 4.4.) A complete picture requires both complementary aspects to be taken into account. In any given situation you can force one feature to be more prominent by virtue of how you choose to interact. If you allow the electrons to travel from source to screen unobserved, their wavelike qualities can emerge, yielding interference. But if you observe the electron en route, you know which path it took, so you'd be at a loss to explain interference. Reality comes to the rescue. Your observation prunes the branches of quantum history. It forces the electron to behave as a particle; since particles go one way *or* the other, no interference pattern forms, so there's nothing to explain.

Nature does weird things. It lives on the edge. But it is careful to bob and weave from the fatal punch of logical paradox.

The Contingency of History

These experiments are remarkable. They provide simple but powerful proof that our world is governed by the quantum laws found by physicists in the twentieth century, and not by the classical laws found by Newton, Maxwell, and Einstein—laws we now recognize as powerful and insightful approximations for describing events at large enough scales. Already we have seen that the quantum laws challenge conventional notions of what happened in the past—those unobserved events that are responsible for what we now see. Some simple variations of these experiments take this challenge to our intuitive notion of how things unfold in time to an even greater, even more surprising level.

The first variation is called the *delayed-choice* experiment and was suggested in 1980 by the eminent physicist John Wheeler. The experiment brushes up against an eerily odd-sounding question: Does the past depend on the future? Note that this is not the same as asking whether we can go back and change the past (a subject we take up in Chapter 15). Instead, Wheeler's experiment, which has been carried out and analyzed in considerable detail, exposes a provocative interplay between events we imagine having taken place in the past, even the distant past, and those we see taking place right now.